



LOS_THEORY ASSIGNMENT - 3

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LINUX FILESYSTEM

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

1. The storage devices available.

- Use lsblk to identify disks and attached storage devices. The -f option also shows filesystem information.

```
lsblk
lsblk -f
df -T
df -h
```

2. The partitions available on each device.

- In lsblk output, entries indented below a disk such as /dev/sda or /dev/nvme0n1 are normally its partitions. Record the actual partition names shown on your machine.

3. The filesystem type used by each partition.

- Use lsblk -f or df -T. Common examples are ext4, vfat, swap, xfs, or other file system types depending on the installation.

4. The device/partition on which / is located.

```
- df -T /
```

The Filesystem column in the output identifies the device or partition mounted at /. Record that exact value.

5. The filesystem type used for /.

- The Type column from df -T / gives the filesystem type used for the root filesystem.

6. The total, used, and available storage space.

- df -h /

The Size, Used, and Avail columns give the total capacity, used space, and available space for the filesystem containing /.

Device/Partition	Filesystem Type	Size	Mount Point
/dev/sda1	ext4	79G	/

Screenshot 1 – Storage devices and filesystems

Run: lsblk -f

```
(kali@kali)-[~]
$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 80.1G 0 disk
└─sda1 8:1    0 80.1G 0 part /

(kali@kali)-[~]
$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
└─sda1 ext4   1.0   root ddf4a404-84cf-4ada-a8f5-a7d36b8335b0 57.7G   21% /

(kali@kali)-[~]
$ df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
udev            devtmpfs   889660         0   889660    0% /dev
tmpfs           tmpfs     197872       1240   196632    1% /run
/dev/sda1       ext4    82083148 17401300 60466300 23% /
tmpfs           tmpfs     989352         4   989348    1% /dev/shm
none            tmpfs      1024         0     1024    0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     989352       272   989080    1% /tmp
none            tmpfs      1024         0     1024    0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197868       116   197752    1% /run/user/1000
none            tmpfs      1024         0     1024    0% /run/credentials/systemd-networkd.service

(kali@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            869M   0 869M   0% /dev
tmpfs           194M  1.3M 193M   1% /run
/dev/sda1       79G   17G  58G  23% /
tmpfs           967M  4.0K 967M   1% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M 272K 966M   1% /tmp
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M 116K 194M   1% /run/user/1000
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
```

Screenshot 2 – Root filesystem and space usage

Run: `df -T / && df -h /`

```
(kali@kali)-[~]
$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 80.1G 0 disk
└─sda1 8:1      0 80.1G 0 part /

(kali@kali)-[~]
$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda   ext4    1.0   root ddf4a404-84cf-4ada-a8f5-a7d36b8335b0  57.7G   21% /

(kali@kali)-[~]
$ df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
udev            devtmpfs   889660         0   889660    0% /dev
tmpfs           tmpfs     197872        1240   196632    1% /run
/dev/sda1       ext4    82083148 17401300 60466300  23% /
tmpfs           tmpfs     989352         4   989348    1% /dev/shm
none            tmpfs      1024         0     1024    0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     989352        272   989080    1% /tmp
none            tmpfs      1024         0     1024    0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197868        116   197752    1% /run/user/1000
none            tmpfs      1024         0     1024    0% /run/credentials/systemd-networkd.service

(kali@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            869M   0 869M   0% /dev
tmpfs           194M  1.3M 193M   1% /run
/dev/sda1       79G   17G  58G  23% /
tmpfs           967M  4.0K 967M   1% /dev/shm
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M 272K 966M   1% /tmp
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M 116K 194M   1% /run/user/1000
none            1.0M   0 1.0M   0% /run/credentials/systemd-networkd.service
```

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

- Use the actual disk and partition names from Exercise 1. A suitable format is:

```
Disk: /dev/sda
|
+-- Partition: /dev/sda1
    |
    +-- Filesystem: ext4
        |
        +-- Mount Point: /
```

Explain in 2–3 sentences the relationship between these four components.

- A disk is the physical or virtual storage device that provides capacity for data. A partition divides that disk into a logical region, a filesystem organizes data inside the partition, and a mount point is the directory through which Linux makes that filesystem accessible.

Screenshot 3 – Actual lsblk layout used for the diagram

Run: `lsblk -f`

```
(kali@kali)-[~]
$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 80.1G 0 disk
└─sda1 8:1      0 80.1G 0 part /

(kali@kali)-[~]
$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda   ext4    1.0   root  ddf4a404-84cf-4ada-a8f5-a7d36b8335b0  57.7G   21% /
└─sda1 ext4    1.0   root  ddf4a404-84cf-4ada-a8f5-a7d36b8335b0  57.7G   21% /

(kali@kali)-[~]
$ df -T
Filesystem      Type      1K-blocks      Used Available Use% Mounted on
udev            devtmpfs   889660          0    889660   0% /dev
tmpfs           tmpfs     197872          1240   196632   1% /run
/dev/sda1       ext4     82083148 17401300 60466300 23% /
tmpfs           tmpfs     989352          4    989348   1% /dev/shm
none            tmpfs      1024           0     1024    0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     989352         272   989080   1% /tmp
none            tmpfs      1024           0     1024    0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197868         116   197752   1% /run/user/1000
none            tmpfs      1024           0     1024    0% /run/credentials/systemd-networkd.service

(kali@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            869M   0 869M   0% /dev
tmpfs           194M  1.3M 193M   1% /run
/dev/sda1       79G   17G  58G  23% /
tmpfs           967M  4.0K 967M   1% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M  272K 966M   1% /tmp
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  116K 194M   1% /run/user/1000
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
```

Exercise 3 – Explore the Root Filesystem

1. Determine your current working directory.

– `pwd`

The `pwd` command prints the absolute path of the current working directory.

2. Navigate to the root directory.

– `cd /`

3. Display the contents of the root directory.

– `ls`

4. Identify all directories directly available under `/`.

– `ls -ld /*/`

Typical Linux systems contain directories such as `/bin`, `/boot`, `/dev`, `/etc`, `/home`, `/lib`, `/media`, `/mnt`, `/opt`, `/proc`, `/root`, `/run`, `/sbin`, `/srv`, `/sys`, `/tmp`, `/usr`, and `/var`. The exact list varies by distribution and installation.

5. Determine whether hidden entries are present.

- `ls -la /`

Entries whose names begin with a dot are hidden by convention. The root directory normally includes `.` and `..` in detailed listings; other hidden entries may or may not be present.

Screenshot 4 – Root directory contents

Run: `pwd && cd / && ls -la`

```
(kali@kali)-[/]
$ pwd
/

(kali@kali)-[/]
$ ls -la
total 976644
drwxr-xr-x 19 root root      4096 Sep  3 00:13 .
drwxr-xr-x 19 root root      4096 Sep  3 00:13 ..
lrwxrwxrwx  1 root root         7 Jun 16 06:14 bin -> usr/bin
drwxr-xr-x  3 root root      4096 Aug 11 10:12 boot
drwxr-xr-x 18 root root     3200 Sep  8 13:38 dev
drwxr-xr-x 186 root root    12288 Sep  7 23:43 etc
drwxr-xr-x  6 root root      4096 Sep  1 00:23 home
lrwxrwxrwx  1 root root      33 Aug 11 10:08 initrd.img -> boot/initrd.img-7.0.12+kali-amd64
lrwxrwxrwx  1 root root      34 Jun 16 11:05 initrd.img.old -> boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx  1 root root         7 Jun 16 06:14 lib -> usr/lib
lrwxrwxrwx  1 root root      9 Jun 16 10:27 lib32 -> usr/lib32
lrwxrwxrwx  1 root root         9 Jun 16 06:14 lib64 -> usr/lib64
drwx----- 2 root root    16384 Jun 16 11:04 lost+found
drwxr-xr-x  2 root root      4096 Jun 16 10:23 media
drwxr-xr-x  2 root root      4096 Jun 16 10:23 mnt
drwxr-xr-x  3 root root      4096 Jun 16 10:27 opt
drwxr-xr-x  2 root root      4096 Sep  3 00:14 ownership_lab
dr-xr-xr-x 313 root root         0 Sep  1 00:16 proc
drwx----- 4 root root      4096 Jun 16 10:25 root
drwxr-xr-x 43 root root      980 Sep  8 13:39 run
lrwxrwxrwx  1 root root         8 Jun 16 06:14/sbin -> usr/sbin
drwxr-xr-x  3 root root      4096 Jun 16 10:29 srv
-rw-----  1 root root 1000000000 Jun 16 11:05 swap
dr-xr-xr-x 13 root root         0 Sep  1 00:16 sys
drwxrwxrwt 13 root root      320 Sep  8 13:41 tmp
drwxr-xr-x 15 root root      4096 Jun 16 10:27 usr
drwxr-xr-x 12 root root      4096 Aug 11 09:29 var
lrwxrwxrwx  1 root root      30 Aug 11 10:08 vmlinuz -> boot/vmlinuz-7.0.12+kali-amd64
lrwxrwxrwx  1 root root      31 Jun 16 11:05 vmlinuz.old -> boot/vmlinuz-6.19.14+kali-amd64

(kali@kali)-[/]
$ ls -d /*/
/bin/  /dev/  /home/  /lib32/  /lost+found/  /mnt/  /ownership_lab/  /root/  /sbin/  /sys/  /usr/
/boot/  /etc/  /lib/  /lib64/  /media/  /opt/  /proc/  /run/  /srv/  /tmp/  /var/
```

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations: `/home`, `/root`, `/etc`, `/var`, `/var/log`, `/tmp`, `/usr`, `/boot`, `/dev`, `/proc`.

- Use `ls -la <directory>` to examine each location. For `/proc` and `/dev`, remember that many entries are virtual or dynamically provided by the kernel rather than ordinary files stored on disk.

Directory	Example Contents	Purpose
<code>/home</code>	User directories, Desktop, Documents (where present)	Stores personal files and user-specific data.
<code>/root</code>	Root user's files and configuration	Home directory of the root/superuser account.

/etc	passwd, group, hostname, hosts	System-wide configuration files.
/var	log, cache, lib, spool	Variable data such as logs, caches and application state.
/var/log	syslog, auth.log, journal, kern.log (varies)	System and application log storage.
/tmp	Temporary files and directories	Temporary working data used by applications and users.
/usr	bin, lib, share, local	Most user-space programs, libraries and shared resources.
/boot	vmlinuz, initrd/initramfs, grub	Bootloader files and Linux kernel-related files.
/dev	null, zero, tty, disks	Device nodes and special device interfaces.
/proc	cpuinfo, meminfo, PID directories	Virtual filesystem exposing kernel and process information.

Example investigation commands:

```
for d in /home /root /etc /var /var/log /tmp /usr /boot /dev /proc; do echo
"===== $d ====="; ls -la "$d" | head -20; done
```

Screenshot 5 – Representative filesystem hierarchy investigation

Run: ls -la /home; ls -la /etc; ls -la /var/log

```
(kali@kali)-[/]
$ ls -la /home
total 24
drwxr-xr-x 6 root root 4096 Sep 1 00:23 .
drwxr-xr-x 19 root root 4096 Sep 3 00:13 ..
drwx----- 5 alice alice 4096 Sep 3 00:34 alice
drwx----- 5 bob bob 4096 Sep 1 00:23 bob
drwx----- 5 charlie charlie 4096 Sep 1 00:23 charlie
drwx--x--x 25 kali kali 4096 Sep 8 13:39 kali

(kali@kali)-[/]
$ ls -la /root
ls: cannot open directory '/root': Permission denied

(kali@kali)-[/]
$ ls -la /etc
total 1640
drwxr-xr-x 186 root root 12288 Sep 7 23:43 .
drwxr-xr-x 19 root root 4096 Sep 3 00:13 ..
-rw-r--r-- 1 root root 3822 Jun 1 01:06 adduser.conf
drwxr-xr-x 2 root root 20480 Aug 11 10:08 alternatives
drwxr-xr-x 8 root root 4096 Jun 16 10:30 apache2
drwxr-xr-x 2 root root 4096 Aug 11 10:07 apparmor
drwxr-xr-x 9 root root 12288 Aug 11 10:10 apparmor.d
drwxr-xr-x 8 root root 4096 Jun 16 11:05 apt
drwxr-xr-x 2 root root 4096 Jun 16 10:28 arp-scan
drwxr-xr-x 3 root root 4096 Jun 16 10:25 avahi
-rw-r--r-- 1 root root 2553 May 3 15:28 bash.bashrc
-rw-r--r-- 1 root root 45 Sep 28 2025 bash_completion
drwxr-xr-x 2 root root 4096 Jun 16 10:23 bash_completion.d
-rw-r--r-- 1 root root 367 May 6 18:25 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Mar 23 09:27 binfmt.d
drwxr-xr-x 2 root root 4096 Aug 11 10:10 bluetooth
drwxr-xr-x 3 root root 4096 Jun 16 10:23 ca-certificates
-rw-r--r-- 1 root root 5287 Jun 16 10:23 ca-certificates.conf
drwxr-s--- 2 root dip 4096 Jun 16 10:25 chatscripts
drwxr-xr-x 2 root root 4096 Aug 11 10:10 chromium
drwxr-xr-x 2 root root 4096 Aug 11 10:10 chromium.d
drwxr-xr-x 2 root root 4096 Jun 16 10:28 cifs-utils
drwxr-xr-x 3 root root 4096 Jun 16 10:23 cloud
drwxr-xr-x 2 colord colord 4096 Aug 11 09:37 colord
```

```

(kali@kali)-[/]
$ ls -la /var
total 52
drwxr-xr-x 12 root root 4096 Aug 11 09:29 .
drwxr-xr-x 19 root root 4096 Sep 3 00:13 ..
drwxr-xr-x 2 root root 4096 Sep 1 00:09 backups
drwxr-xr-x 19 root root 4096 Aug 11 09:37 cache
drwxr-xr-x 77 root root 4096 Aug 11 10:12 lib
drwxr-xr-x 2 root root 4096 Jun 16 06:14 local
lrwxrwxrwx 1 root root 9 Jun 16 10:23 lock -> /run/lock
drwxr-xr-x 21 root root 4096 Sep 8 13:39 log
drwxrwsr-x 2 root mail 4096 Jun 16 10:23 mail
drwxr-xr-x 2 root root 4096 Jun 16 10:23 opt
lrwxrwxrwx 1 root root 4 Jun 16 10:23 run -> /run
drwxr-xr-x 3 root root 4096 Jun 16 10:23 spool
drwxrwxrwt 8 root root 4096 Sep 8 13:39 tmp
-rw-r--r-- 1 root root 255 Jun 16 10:27 .updated
drwxr-xr-x 3 root root 4096 Jun 16 10:26 www

(kali@kali)-[/]
$ ls -la /var/log
total 2528
drwxr-xr-x 21 root root 4096 Sep 8 13:39 .
drwxr-xr-x 12 root root 4096 Aug 11 09:29 ..
-rw-r--r-- 1 root root 0 Sep 1 00:08 alternatives.log
-rw-r--r-- 1 root root 111418 Aug 11 10:10 alternatives.log.1
drwxr-xr-x 2 root adm 4096 Jun 16 10:30 apache2
drwxr-xr-x 2 root root 4096 Sep 1 00:08 apt
-rw-r--r-- 1 root root 0 Sep 7 23:43 boot.log
-rw-r--r-- 1 root root 15421 Sep 7 23:43 boot.log.1
-rw-r--r-- 1 root root 6927 Aug 27 00:54 boot.log.2
-rw-r--r-- 1 root root 16236 Aug 24 12:18 boot.log.3
-rw-r--r-- 1 root root 15082 Aug 21 14:08 boot.log.4
-rw-r--r-- 1 root root 68392 Aug 16 04:55 boot.log.5
-rw-rw-r-- 1 root utmp 0 Sep 1 00:08 btmp
-rw-rw-r-- 1 root utmp 384 Aug 22 10:50 btmp.1
-rw-r--r-- 1 root root 0 Sep 1 00:08 dpkg.log
-rw-r--r-- 1 root root 2026079 Aug 27 00:18 dpkg.log.1
-rw-r--r-- 1 root root 8409 Aug 11 10:12 fontconfig.log
drwxr-xr-x 2 _gvm _gvm 4096 Apr 28 09:56 gvm
drwxr-xr-x 3 inetsim inetsim 4096 Jun 16 10:29 inetsim
drwxr-sr-x+ 3 root systemd-journal 4096 Aug 11 09:29 journal

```

```

(kali@kali)-[/]
$ ls -la /tmp
total 12
drwxrwxrwt 13 root root 320 Sep 8 13:41 .
drwxr-xr-x 19 root root 4096 Sep 3 00:13 ..
-rw-r--r-- 1 kali kali 0 Sep 1 00:17 config-err-0TKYgZ
drwxrwxrwt 2 root root 40 Sep 1 00:16 .font-unix
drwxrwxrwt 2 root root 60 Sep 1 00:17 .ICE-unix
drwxr-xr-x 3 root root 60 Sep 1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-colord.service
-bMcddDD
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-ModemManager.s
ervice-bxfVGP
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-polkit.service
-UYJ3xl
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-systemd-logind
.service-DYQGuj
drwxr-xr-x 3 root root 60 Sep 1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-upower.service
-xLbchu
drwxrwxrwt 2 root root 40 Sep 1 00:16 VMwareDnD
drwxr-xr-x 2 root root 40 Sep 1 00:16 vmware-root_464-826976206
-r--r--r-- 1 root root 11 Sep 1 00:17 .X0-lock
drwxrwxrwt 2 root root 60 Sep 1 00:17 .X11-unix
-rw-r--r-- 1 kali kali 398 Sep 1 00:17 .xfsm-ICE-HVV0U3
drwxrwxrwt 2 root root 40 Sep 1 00:16 .XIM-unix

```

```

(kali@kali)-[/]
$ ls -la /usr
total 228
drwxr-xr-x 15 root root 4096 Jun 16 10:27 .
drwxr-xr-x 19 root root 4096 Sep 3 00:13 ..
drwxr-xr-x 2 root root 131072 Aug 27 00:18 bin
drwxr-xr-x 2 root root 4096 Jun 16 06:14 games
drwxr-xr-x 5 root root 4096 Jun 16 10:26 i686-w64-mingw32
drwxr-xr-x 85 root root 20480 Aug 11 10:00 include
drwxr-xr-x 112 root root 4096 Aug 11 10:00 lib
drwxr-xr-x 3 root root 4096 Aug 11 09:58 lib32
drwxr-xr-x 2 root root 4096 Aug 11 09:58 lib64
drwxr-xr-x 27 root root 4096 Aug 11 10:06 libexec
drwxr-xr-x 11 root root 4096 Jun 16 10:23 local
drwxr-xr-x 2 root root 20480 Aug 11 10:06 sbin
drwxr-xr-x 360 root root 12288 Aug 27 00:18 share
drwxr-xr-x 3 root root 4096 Jun 16 10:27 src

```

```

(kali@kali)-[/]
$ ls -la /boot
total 408324
drwxr-xr-x 3 root root      4096 Aug 11 10:12 .
drwxr-xr-x 19 root root      4096 Sep  3 00:13 ..
-rw-r--r-- 1 root root 296567 May  5 02:54 config-6.19.14+kali-amd64
-rw-r--r-- 1 root root 297482 Jun 18 05:48 config-7.0.12+kali-amd64
drwxr-xr-x 6 root root      4096 Aug 11 10:09 grub
-rw-r--r-- 1 root root 193633189 Jun 16 11:05 initrd.img-6.19.14+kali-amd64
-rw-r--r-- 1 root root 195855355 Aug 11 10:12 initrd.img-7.0.12+kali-amd64
-rw-r--r-- 1 root root      92 May  5 02:54 System.map-6.19.14+kali-amd64
-rw-r--r-- 1 root root      91 Jun 18 05:48 System.map-7.0.12+kali-amd64
-rw-r--r-- 1 root root 13931008 May  5 02:54 vmlinuz-6.19.14+kali-amd64
-rw-r--r-- 1 root root 14070272 Jun 18 05:48 vmlinuz-7.0.12+kali-amd64

```

```

(kali@kali)-[/]
$ ls -la /dev
total 4
drwxr-xr-x 18 root root      3200 Sep  8 13:38 .
drwxr-xr-x 19 root root      4096 Sep  3 00:13 ..
crw-r--r-- 1 root root      10, 235 Sep  1 00:17 autofs
drwxr-xr-x 2 root root        80 Sep  1 00:17 block
drwxr-xr-x 2 root root        60 Sep  1 00:16 bsg
crw----- 1 root root      10, 234 Sep  1 00:16 btrfs-control
drwxr-xr-x 3 root root        60 Sep  1 00:16 bus
drwxr-xr-x 2 root root     3120 Sep  8 13:40 char
crw----- 1 root root        5,  1 Sep  1 00:16 console
lrwxrwxrwx 1 root root        11 Sep  1 00:16 core -> /proc/kcore
crw----- 1 root root      10, 259 Sep  1 00:17 cpu_dma_latency
crw----- 1 root root      10, 203 Sep  1 00:16 cuse
drwxr-xr-x 7 root root       140 Sep  1 00:16 disk
drwxr-xr-x 2 root root        60 Sep  1 00:16 dma_heap
crw-rw----+ 1 root audio    14,  9 Sep  1 00:17 dmmidi
drwxr-xr-x 3 root root       100 Sep  1 00:17 dri
crw-rw---- 1 root video    29,  0 Sep  1 00:17 fb0
lrwxrwxrwx 1 root root       13 Sep  1 00:16 fd -> /proc/self/fd
crw-rw-rw- 1 root root        1,  7 Sep  1 00:17 full
crw-rw-rw- 1 root root      10, 229 Sep  1 00:17 fuse
crw----- 1 root root     244,  0 Sep  1 00:17 hidraw0
crw----- 1 root root     244,  1 Sep  8 13:39 hidraw1
crw----- 1 root root      10, 228 Sep  1 00:17 hpet
drwxr-xr-x 2 root root        0 Sep  1 00:16 hugepages

```

```

(kali@kali)-[/]
$ ls -la /proc
total 4
dr-xr-xr-x 298 root      root      0 Sep  1 00:16 .
drwxr-xr-x 19 root      root      4096 Sep  3 00:13 ..
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 1
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 10
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 100
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 101
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 102
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 103
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 104
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 105
dr-xr-xr-x 9 systemd-network systemd-network 0 Sep  1 00:40 10524
dr-xr-xr-x 9 root      root      0 Sep  1 00:46 10547
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 106
dr-xr-xr-x 9 root      root      0 Sep  3 01:33 106682
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 107
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 108
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 110
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 111
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 112
dr-xr-xr-x 9 root      root      0 Sep  1 00:17 1140
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1164
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1167
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1187
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1188
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 119
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1190
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1191
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1192
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1193
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1194
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 12
dr-xr-xr-x 9 root      root      0 Sep  1 00:16 120
dr-xr-xr-x 9 root      root      0 Sep  6 02:36 120197
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1205
dr-xr-xr-x 9 root      root      0 Sep  6 02:39 120909
dr-xr-xr-x 9 kali      kali      0 Sep  6 02:39 121036
dr-xr-xr-x 9 kali      kali      0 Sep  1 00:17 1284

```

Exercise 5 – Find Your Home Directory

1. Determine your username.

```
- whoami
```

2. Determine your home directory.

```
- echo "$HOME"
```

```
getent passwd "$(whoami) "
```

The HOME environment variable normally identifies the current user's home directory. The getent command can show the account entry, where the home directory is one of the colon-separated fields.

3. Navigate to your home directory.

```
- cd "$HOME"
```

4. Display its absolute path.

```
- pwd
```

5. List its normal contents.

```
- ls
```

6. List all contents, including hidden files.

```
- ls -la
```

Screenshot 6 – Home directory discovery and listing

Run: whoami; echo "\$HOME"; cd "\$HOME"; pwd; ls -la

```
(kali@kali)-[/]
$ whoami
kali

(kali@kali)-[/]
$ echo "$HOME"
/home/kali

(kali@kali)-[/]
$ getent passwd "$(whoami)"
kali:x:1000:1000::/home/kali:/usr/bin/zsh

(kali@kali)-[/]
$ cd "$HOME"

(kali@kali)-[~]
$ pwd
/home/kali

(kali@kali)-[~]
$ ls
1.out  Documents  exitstat.sh  filters  Music  practice  Public  testdir  test.sh  Videos
Desktop  Downloads  file1.txt   Linux   Pictures  Projects  Templates  testdirr  Test.txt

(kali@kali)-[~]
$ ls -la
total 200
drwx--x--x 25 kali kali 4096 Sep  8 13:39 .
drwxr-xr-x  6 root root 4096 Sep  1 00:23 ..
-rw-rw-r--  1 kali kali  29 Aug 26 23:53 1.out
-rw-rw-r--  1 kali kali 3987 Sep  1 00:11 .bash_history
-rw-r--r--  1 kali kali  220 Jun 16 10:30 .bash_logout
-rw-r--r--  1 kali kali 5578 Jun 16 10:30 .bashrc
-rw-r--r--  1 kali kali 3526 Jun 16 10:30 .bashrc.original
drwxrwxr-x 11 kali kali 4096 Aug 13 14:23 .cache
drwxr-xr-x 17 kali kali 4096 Sep  1 03:02 .config
drwxr-xr-x  2 kali kali 4096 Aug 11 09:37 Desktop
-rw-r--r--  1 kali kali  35 Aug 11 09:37 .dmrc
drwxr-xr-x  3 kali kali 4096 Sep  1 02:56 Documents
drwxr-xr-x  2 kali kali 4096 Aug 11 09:37 Downloads
-rwxrwxr-x  1 kali kali  108 Aug 12 14:39 exitstat.sh
```

Exercise 6 – Hidden Files Investigation

Examine your home directory. Compare: `ls` and `ls -a`.

```
- ls
ls -a
```

Use `ls -la` to identify at least two hidden entries. Common examples include configuration directories such as `.config` and shell configuration files such as `.bashrc`, but the exact entries depend on the distribution and desktop environment.

Question	Answer
1. How does Linux identify a hidden file?	A filename is conventionally hidden when it begins with a dot (.).
2. Why does a normal <code>ls</code> not display these files?	<code>ls</code> hides dot-prefixed entries by default to keep ordinary directory listings uncluttered.
3. What do <code>.</code> and <code>..</code> mean?	<code>.</code> means the current directory; <code>..</code> means the parent directory.
4. Can a hidden file still be accessed if its name is known?	Yes. It can be accessed normally by specifying its path/name, for example <code>cat ~/.bashrc</code> .
5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?	Hidden status is only a naming convention. Legitimate configuration files and potentially suspicious files can both be hidden, so their contents, ownership, permissions, timestamps and context should be examined.

Screenshot 7 – Comparison of `ls` and `ls -a`

Run: `ls`; `ls -a`

```
(kali@kali)-[~]
└─$ ls
Desktop  Documents  exitstat.sh  filters  Music  practice  Public  testdir  test.sh  Videos
└─$ ls -a
.         Desktop  .gnupg      .profile  Test.txt
..        .dmrc     .ICEauthority .Projects Videos
1.out     Documents .java       Public    .viminfo
Downloads Downloads Linux        .ssh      .Xauthority
exitstat.sh .local   .sudo_as_admin_successful .xsession-errors
.bashrc    .face    .mozilla    Templates .xsession-errors.old
.bashrc.original .face.icon Music      testdir   .zprofile
.cache     file1.txt Pictures    testdirr  .zsh_history
.config    filters  practice    test.sh   .zshrc

(kali@kali)-[~]
└─$ ls -la
total 200
drwx--x--x 25 kali kali 4096 Sep  8 13:39 .
drwxr-xr-x  6 root root 4096 Sep  1 00:23 ..
-rw-rw-r--  1 kali kali   29 Aug 26 23:53 1.out
-rw-----  1 kali kali 3987 Sep  1 00:11 .bash_history
-rw-r--r--  1 kali kali  220 Jun 16 10:30 .bash_logout
-rw-r--r--  1 kali kali 5578 Jun 16 10:30 .bashrc
-rw-r--r--  1 kali kali 3526 Jun 16 10:30 .bashrc.original
drwxrwxr-x 11 kali kali 4096 Aug 13 14:23 .cache
drwxr-xr-x 17 kali kali 4096 Sep  1 03:02 .config
drwxr-xr-x  2 kali kali 4096 Aug 11 09:37 Desktop
-rw-r--r--  1 kali kali   35 Aug 11 09:37 .dmrc
drwxr-xr-x  3 kali kali 4096 Sep  1 02:56 Documents
drwxr-xr-x  2 kali kali 4096 Aug 11 09:37 Downloads
-rwxrwxr-x  1 kali kali  108 Aug 12 14:39 exitstat.sh
-rw-r--r--  1 kali kali 11759 Jun 16 10:30 .face
lrwxrwxrwx  1 kali root    5 Jun 16 10:30 .face.icon → .face
-rw-rw-r--  1 kali kali    0 Aug 24 12:22 file1.txt
drwxrwxr-x  2 kali kali 4096 Aug 24 14:44 filters
drwx-----  3 kali kali 4096 Aug 11 09:37 .gnupg
-rw-----  1 kali kali    0 Aug 11 09:37 .ICEauthority
drwxr-xr-x  3 kali kali 4096 Jun 16 10:30 .java
```

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: mkdir

Create the following hierarchy and files.

- The directories can be created in one command using mkdir -p:

```
mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs  
cyber_lab/scripts cyber_lab/reports
```

Create the required files:

```
touch cyber_lab/evidence/network/traffic.txt \  
      cyber_lab/evidence/system/processes.txt \  
      cyber_lab/logs/security.log \  
      cyber_lab/scripts/check.sh \  
      cyber_lab/reports/investigation.txt
```

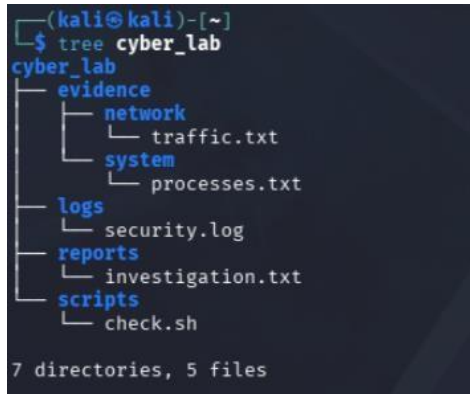
Verify that the required hierarchy has been created successfully.

- tree cyber_lab

The tree command gives a visual hierarchy.

Screenshot 8 – Created cyber_lab hierarchy

Run: tree cyber_lab



```
(kali@kali)-[~]  
$ tree cyber_lab  
cyber_lab  
├── evidence  
│   ├── network  
│   │   └── traffic.txt  
│   └── system  
│       └── processes.txt  
├── logs  
│   └── security.log  
├── reports  
│   └── investigation.txt  
└── scripts  
    └── check.sh  
  
7 directories, 5 files
```

Exercise 8 – Absolute and Relative Paths

Assume your current directory is: ~/cyber_lab

1. Navigate to evidence/network using a relative path.

```
- cd evidence/network  
pwd
```

2. Navigate to /etc using an absolute path.

```
- cd /etc  
pwd
```

3. Return to your home directory using the shortest possible command.

- cd ~
pwd

4. Navigate to cyber_lab/reports.

- cd ~/cyber_lab/reports
pwd

5. Move one directory upward.

- cd ..
pwd

6. Display the current directory after every operation.

- The pwd command should be run after each cd operation. The sequence above demonstrates the required navigation.

Question	Answer
What is an absolute path?	A path that starts at the root directory / and specifies the complete location, such as /etc/hosts.
What is a relative path?	A path interpreted from the current working directory, such as evidence/network.
What is the difference between / and ~?	/ is the root of the entire Linux file system; ~ is shorthand for the current user's home directory.

Screenshot 8b – Absolute and relative path navigation

Run: cd/pwd sequence through evidence/network, /etc, ~, cyber_lab/reports, and one level up

```
(kali@kali)-[/etc]
$ cd ~/cyber_lab
(kali@kali)-[~/cyber_lab]
$ pwd
/home/kali/cyber_lab
(kali@kali)-[~/cyber_lab]
$ cd evidence/network
(kali@kali)-[~/cyber_lab/evidence/network]
$ pwd
/home/kali/cyber_lab/evidence/network
(kali@kali)-[~/cyber_lab/evidence/network]
$ cd /etc
(kali@kali)-[/etc]
$ pwd
/etc
(kali@kali)-[/etc]
$ cd ~
(kali@kali)-[~]
$ pwd
/home/kali
(kali@kali)-[~]
$ cd ~/cyber_lab/reports
(kali@kali)-[~/cyber_lab/reports]
$ pwd
/home/kali/cyber_lab/reports
(kali@kali)-[~/cyber_lab/reports]
$ cd ..
(kali@kali)-[~/cyber_lab]
$ pwd
/home/kali/cyber_lab
```

Exercise 9 – Identify File Types

Investigate `/etc/passwd`, `/etc`, `/bin/bash`, `/dev/null`, `/proc` and the files created in Exercise 7.

```
file /etc/passwd
file /etc
file /bin/bash
file /dev/null
file /proc
```

```
ls -l /etc/passwd /etc /bin/bash /dev/null /proc
```

```
file ~/cyber_lab/logs/security.log ~/cyber_lab/scripts/check.sh
```

Expected classifications are shown below. Exact file permissions, ownership and symlink targets can vary.

Object	Type
<code>/etc/passwd</code>	Regular text file containing account database information
<code>/etc</code>	Directory
<code>/bin/bash</code>	Executable file; on many modern distributions <code>/bin</code> may be a symbolic link to <code>/usr/bin</code>
<code>/dev/null</code>	Character special device
<code>/proc</code>	Virtual/pseudo-filesystem directory
<code>security.log</code>	Regular file (text file after text is written)
<code>check.sh</code>	Regular file; it becomes an executable program only if executable permissions are set and it contains a valid script

Screenshot 9 – file and ls -l investigation

Run: `file /etc/passwd /etc /bin/bash /dev/null /proc`

```
(kali@kali)-[~/cyber_lab]
$ file /etc/passwd
/etc/passwd: ASCII text

(kali@kali)-[~/cyber_lab]
$ file /etc
/etc: directory

(kali@kali)-[~/cyber_lab]
$ file /bin/bash
/bin/bash: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=e1983fd58f88f72cd98b92c28692a9eef51d8da1, for GNU/Linux 3.2.0, stripped

(kali@kali)-[~/cyber_lab]
$ file /dev/null
/dev/null: character special (1/3)

(kali@kali)-[~/cyber_lab]
$ file /proc
/proc: directory

(kali@kali)-[~/cyber_lab]
$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3373 Sep  1 00:23 /etc/passwd

(kali@kali)-[~/cyber_lab]
$ ls -ld /etc
drwxr-xr-x 186 root root 12288 Sep  7 23:43 /etc

(kali@kali)-[~/cyber_lab]
$ ls -l /bin/bash
-rwxr-xr-x 1 root root 1384752 May  3 15:28 /bin/bash

(kali@kali)-[~/cyber_lab]
$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Sep  1 00:17 /dev/null

(kali@kali)-[~/cyber_lab]
$ ls -ld /proc
dr-xr-xr-x 299 root root 0 Sep  1 00:16 /proc
```

```
(kali@kali)-[~/cyber_lab]
$ ls -ld /proc
dr-xr-xr-x 299 root root 0 Sep  1 00:16 /proc

(kali@kali)-[~/cyber_lab]
$ file ~/cyber_lab/logs/security.log
/home/kali/cyber_lab/logs/security.log: empty

(kali@kali)-[~/cyber_lab]
$ file ~/cyber_lab/scripts/check.sh
/home/kali/cyber_lab/scripts/check.sh: empty

(kali@kali)-[~/cyber_lab]
$ ls -l ~/cyber_lab/logs/security.log
-rw-rw-r-- 1 kali kali 0 Sep  8 13:52 /home/kali/cyber_lab/logs/security.log

(kali@kali)-[~/cyber_lab]
$ ls -l ~/cyber_lab/scripts/check.sh
-rw-rw-r-- 1 kali kali 0 Sep  8 13:52 /home/kali/cyber_lab/scripts/check.sh
```

Exercise 10 – Examine File Metadata

Use: ~/cyber_lab/reports/investigation.txt

Add a small amount of text to the file.

```
echo "Filesystem investigation started." >>
~/cyber_lab/reports/investigation.txt
```

Examine it using **ls -l** and **stat investigation.txt**.

```
cd ~/cyber_lab/reports
ls -l investigation.txt
stat investigation.txt
```

Metadata Item	How to identify it
Filename	The Name field in stat or the filename displayed by ls.
File size	Size field in stat; bytes are shown.
File type	File type shown by stat; file investigation.txt is also useful.
Owner	Uid/User field in stat or owner column in ls -l.
Group	Gid/Group field in stat or group column in ls -l.
Inode number	Inode field in stat.
Access time	Access field in stat.
Modification time	Modify field in stat.
Change time	Change field in stat; this is metadata/status change time on Linux, not file creation time.

Question	Answer
1. What is metadata?	Metadata is information describing a file, such as its size, owner, permissions, inode and timestamps.
2. Is metadata the same as file content?	No. File content is the data stored in the file, while metadata describes the file and how it is managed by the filesystem.
3. Which timestamp changes when file contents are modified?	The modification time (mtime) changes when the file's contents are modified. Access time and change time may

	also change depending on the operation and filesystem settings.
--	---

Screenshot 10 – File metadata using `ls -l` and `stat`

Run: `ls -l investigation.txt && stat investigation.txt`

```
(kali㉿kali)-[~/cyber_lab]
$ echo "Filesystem investigation started." >> ~/cyber_lab/reports/investigation.txt

(kali㉿kali)-[~/cyber_lab]
$ cd ~/cyber_lab/reports

(kali㉿kali)-[~/cyber_lab/reports]
$ ls -l investigation.txt
-rw-rw-r-- 1 kali kali 34 Sep  8 14:02 investigation.txt

(kali㉿kali)-[~/cyber_lab/reports]
$ stat investigation.txt
  File: investigation.txt
  Size: 34          Blocks: 8          IO Block: 4096   regular file
Device: 8,1      Inode: 787868      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-09-08 13:52:52.377807155 -0400
Modify: 2026-09-08 14:02:26.090064847 -0400
Change: 2026-09-08 14:02:26.090064847 -0400
 Birth: 2026-09-08 13:52:52.377807155 -0400
```

Exercise 11 – Explore /etc

Investigate /etc and locate passwd, group, hostname, hosts.

```
ls -l /etc/passwd /etc/group /etc/hostname /etc/hosts
cat /etc/passwd
cat /etc/group
cat /etc/hostname
cat /etc/hosts
```

1. What kind of information is primarily stored under /etc?

/etc primarily contains system-wide configuration files and configuration directories used by the operating system and installed services.

2. What information does /etc/passwd contain?

It contains one account record per line with fields such as username, password placeholder (commonly x), UID, GID, descriptive information, home directory, and login shell. Password hashes are normally stored separately in /etc/shadow, which has restricted permissions.

3. What information does /etc/hostname contain?

It contains the configured hostname of the Linux system, normally as a short text value.

4. Why is /etc security-sensitive?

Many files under /etc control authentication, users, services, networking, scheduled tasks and system behavior. Unauthorized changes can therefore weaken security or alter system operation.

Screenshot 11 – /etc files and permitted contents

Run: ls -l /etc/passwd /etc/group /etc/hostname /etc/hosts

```
(kali@kali)-[~/cyber_lab/reports]
$ ls -l /etc/passwd /etc/group /etc/hostname /etc/hosts
-rw-r--r-- 1 root root 1409 Sep  3 01:44 /etc/group
-r--r--r-- 1 root root   5 Jun 16 10:24 /etc/hostname
-rw-r--r-- 1 root root 124 Jun 16 10:30 /etc/hosts
-rw-r--r-- 1 root root 3373 Sep  1 00:23 /etc/passwd

(kali@kali)-[~/cyber_lab/reports]
$ cat /etc/hostname
kali

(kali@kali)-[~/cyber_lab/reports]
$ cat /etc/hosts
127.0.0.1    localhost
127.0.1.1    kali
::1         localhost ip6-localhost ip6-loopback
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
```

Exercise 12 – Explore Linux Log Storage

Investigate /var/log.

```
ls -lah /var/log
find /var/log -maxdepth 2 -type f -printf '%TY-%Tm-%Td %TH:%TM %p\n'
2>/dev/null | sort -r | head -20
find /var/log -maxdepth 1 -type d -print
```

Available logs vary by distribution. Depending on the system, you may see files or directories associated with authentication, system messages, package management, kernel messages, boot activity, or systemd journal data.

Identify available log files, directories, recently modified entries, readable entries, and entries you cannot read as a normal user.

Use `ls -lah /var/log` and the `find` command above. Try reading only files you are permitted to access. If a permission-denied message appears, record that observation rather than changing permissions.

Question	Answer
1. Why does Linux maintain logs?	Logs record system, service, authentication, kernel and application events so that administrators can monitor and troubleshoot the system.

2. Why are logs useful during incident investigation?	They can provide a timeline of logins, service activity, errors, process/service events and other indicators that help reconstruct suspicious activity.
3. Why might access to some logs be restricted?	Logs may contain sensitive information such as usernames, authentication events, system details or application data, so permissions restrict access to authorized users.

Screenshot 12 – /var/log investigation

Run: `ls -lah /var/log`

```
(kali㉿kali)-[~/cyber_lab/reports]
$ ls -lah /var/log
total 2.5M
drwxr-xr-x 21 root      root      4.0K Sep  8 13:39 .
drwxr-xr-x 12 root      root      4.0K Aug 11 09:29 ..
-rw-r--r--  1 root      root         0 Sep  1 00:08 alternatives.log
-rw-r--r--  1 root      root    109K Aug 11 10:10 alternatives.log.1
drwxr-x-  2 root      adm       4.0K Jun 16 10:30 apache2
drwxr-xr-x  2 root      root      4.0K Sep  1 00:08 apt
-rw-  1 root      root         0 Sep  7 23:43 boot.log
-rw-  1 root      root    16K Sep  7 23:43 boot.log.1
-rw-  1 root      root    6.8K Aug 27 00:54 boot.log.2
-rw-  1 root      root    16K Aug 24 12:18 boot.log.3
-rw-  1 root      root    15K Aug 21 14:08 boot.log.4
-rw-  1 root      root    67K Aug 16 04:55 boot.log.5
-rw-rw-  1 root      utmp         0 Sep  1 00:08 btmp
-rw-rw-  1 root      utmp    384 Aug 22 10:50 btmp.1
-rw-r--r--  1 root      root         0 Sep  1 00:08 dpkg.log
-rw-r--r--  1 root      root    2.0M Aug 27 00:18 dpkg.log.1
-rw-r--r--  1 root      root    8.3K Aug 11 10:12 fontconfig.log
drwxr-xr-x  2 _gvm      _gvm       4.0K Apr 28 09:56 gvm
drwx-  3 inetsim      inetsim     4.0K Jun 16 10:29 inetsim
drwxr-sr-x+ 3 root      systemd-journal 4.0K Aug 11 09:29 journal
-rw-rw-r--  1 root      utmp         0 Jun 16 10:23 lastlog
drwx--x--x  2 root      root      4.0K Sep  1 00:17 lightdm
-rw-r--r--  1 root      root     59 Sep  1 00:16 macchanger.log.1.gz
-rw-r--r--  1 root      root     59 Aug 24 12:30 macchanger.log.2.gz
-rw-r--r--  1 root      root     61 Aug 24 12:10 macchanger.log.3.gz
-rw-r--r--  1 root      root     62 Aug 15 09:02 macchanger.log.4.gz
drwxr-  2 mosquitto   mosquitto  4.0K May 14 05:43 mosquitto
drwxr-s-  2 mysql      adm       4.0K Jun 16 10:29 mysql
drwxr-xr-x  2 root      adm       4.0K Jun 16 10:28 nginx
drwxr-xr-x  2 _gvm      _gvm      4.0K Aug 11 10:10 notus-scanner
drwxr-xr-x  2 root      root      4.0K Apr 27 17:25 openvpn
drwxrwxr-t  2 root      postgres  4.0K Aug 16 04:55 postgresql
drwx-  2 root      root      4.0K Jun 16 10:23 private
drwxr-s-  2 redis      adm       4.0K Jun 16 10:29 redis
drwxr-xr-x  4 root      root      4.0K Jun 16 10:25 runit
drwxr-x-  2 root      adm       4.0K May 26 08:46 samba
drwx-  2 speech-dispatcher root      4.0K Jun 16 10:25 speech-dispatcher
drwxr-xr-x  2 stunnel   stunnel    4.0K Jun 16 10:28 stunnel
```

```
(kali@kali)~/cyber_lab/reports]
$ find /var/log -maxdepth 1 -type f -printf '%TY-%Tm-%Td %TH:%TM %p\n' 2>/dev/null | sort -r | head -20
2026-09-08 13:40 /var/log/Xorg.0.log
2026-09-08 13:39 /var/log/vmware-vmtoolsd-kali.log
2026-09-08 13:39 /var/log/vmware-network.log
2026-09-07 23:43 /var/log/vmware-network.1.log
2026-09-07 23:43 /var/log/boot.log.1
2026-09-07 23:43 /var/log/boot.log
2026-09-06 02:36 /var/log/vmware-network.2.log
2026-09-03 01:45 /var/log/wtmp.db
2026-09-03 00:10 /var/log/vmware-network.3.log
2026-09-01 00:40 /var/log/vmware-network.4.log
2026-09-01 00:17 /var/log/wtmp
2026-09-01 00:17 /var/log/vmware-vmtoolsd-kali.log
2026-09-01 00:17 /var/log/vmware-vmtoolsd-root.log
2026-09-01 00:17 /var/log/vmware-network.5.log
2026-09-01 00:16 /var/log/vmware-vmtoolsd-root.log
2026-09-01 00:16 /var/log/macchanger.log.1.gz
2026-09-01 00:15 /var/log/Xorg.0.log.old
2026-09-01 00:15 /var/log/vmware-vmtoolsd-root.1.log
2026-09-01 00:15 /var/log/vmware-network.6.log
2026-09-01 00:14 /var/log/vmware-network.8.log

(kali@kali)~/cyber_lab/reports]
$ find /var/log -maxdepth 1 -type d -print
/var/log
/var/log/samba
/var/log/mysql
/var/log/runit
/var/log/nginx
/var/log/apt
/var/log/mosquitto
/var/log/postgresql
/var/log/inetsim
/var/log/private
/var/log/sysstat
/var/log/stunnel
/var/log/gvm
/var/log/lightdm
/var/log/apache2
/var/log/journal
```

Exercise 13 – Investigate /tmp

Explore /tmp without modifying files belonging to other users.

```
ls -lah /tmp
find /tmp -maxdepth 1 -mindepth 1 -printf '%TY-%Tm-%Td %TH:%TM %u %p\n'
2>/dev/null | sort -r | head -20
```

The commands show ordinary and hidden entries, ownership, and recently modified entries. Do not delete, edit or change permissions of files that you do not own.

1. Whether files/directories currently exist.

Use `ls -lah /tmp` and record the entries actually present on your system.

2. Whether hidden files exist.

The `-a` option in `ls` displays dot-prefixed entries.

3. Who owns some of the entries.

The owner column in `ls -l` or the `%u` field in the `find` command identifies the owner.

4. Which entries were modified recently.

The `find` command sorts entries by modification timestamp so the newest entries appear first.

Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories can contain transient files created by applications, installers, scripts and users. Suspicious programs may also place downloaded payloads, scripts, staging files or other

short-lived artifacts there, so timestamps, ownership and contents can be useful investigative clues.

Screenshot 13 – /tmp ownership and recent entries

Run: `ls -lah /tmp`

```
(kali@kali) [~/cyber_lab/reports]
$ ls -lah /tmp
total 12K
drwxrwxrwt 13 root root 320 Sep  8 14:06 .
drwxr-xr-x 19 root root 4.0K Sep  3 00:13 ..
-rw----- 1 kali kali  0 Sep  1 00:17 config-err-0TKYgZ
drwxrwxrwt 2 root root  40 Sep  1 00:16 .font-unix
drwxrwxrwt 2 root root  60 Sep  1 00:17 .ICE-unix
drwx----- 3 root root  60 Sep  1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-colord.service-bMcdDD
drwx----- 3 root root  60 Sep  1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-ModemManager.service-bxfVGP
drwx----- 3 root root  60 Sep  1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-polkit.service-UYJ3xL
drwx----- 3 root root  60 Sep  1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-systemd-logind.service-DYQGuj
drwx----- 3 root root  60 Sep  1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-upower.service-xLbchu
drwxrwxrwt 2 root root  40 Sep  1 00:16 VMwareDnD
drwx----- 2 root root  40 Sep  1 00:16 vmware-root_464-826976206
-r--r--r-- 1 root root  11 Sep  1 00:17 .X0-lock
drwxrwxrwt 2 root root  60 Sep  1 00:17 .X11-unix
-rw----- 1 kali kali 398 Sep  1 00:17 .xfsm-ICE-HVV0U3
drwxrwxrwt 2 root root  40 Sep  1 00:16 .XIM-unix

(kali@kali) [~/cyber_lab/reports]
$ find /tmp -maxdepth 1 -mindepth 1 -printf '%TY-%Tm-%Td %TH:%TM %u %p\n' 2>/dev/null | sort -r | head
-20
2026-09-01 00:17 root /tmp/.X11-unix
2026-09-01 00:17 root /tmp/.X0-lock
2026-09-01 00:17 root /tmp/systemd-private-45a3d694897f445c82c735328e937a09-upower.service-xLbchu
2026-09-01 00:17 root /tmp/systemd-private-45a3d694897f445c82c735328e937a09-colord.service-bMcdDD
2026-09-01 00:17 root /tmp/.ICE-unix
2026-09-01 00:17 kali /tmp/.xfsm-ICE-HVV0U3
2026-09-01 00:17 kali /tmp/config-err-0TKYgZ
2026-09-01 00:16 root /tmp/.XIM-unix
2026-09-01 00:16 root /tmp/vmware-root_464-826976206
2026-09-01 00:16 root /tmp/VMwareDnD
2026-09-01 00:16 root /tmp/systemd-private-45a3d694897f445c82c735328e937a09-systemd-logind.service-DYQGuj
2026-09-01 00:16 root /tmp/systemd-private-45a3d694897f445c82c735328e937a09-polkit.service-UYJ3xL
2026-09-01 00:16 root /tmp/systemd-private-45a3d694897f445c82c735328e937a09-ModemManager.service-bxfVGP
2026-09-01 00:16 root /tmp/.font-unix
```

Exercise 14 – Explore the /proc Filesystem

Investigate /proc.

```
cat /proc/cpuinfo
cat /proc/meminfo
echo $$
ls -ld /proc/$$
```

1. What information does /proc/cpuinfo provide?

It provides kernel-exposed information about the processor, such as processor entries, model information, CPU capabilities, clock-related information and other architecture details.

2. What information does /proc/meminfo provide?

It provides memory statistics such as total memory, free memory, available memory, buffers, cache and swap information.

3. Why do directories with numbers appear inside /proc?

Numeric directories normally represent process IDs (PIDs). Each process can have a corresponding /proc/<PID> directory containing information about that process.

4. Is /proc an ordinary disk directory?

No. /proc is a virtual/pseudo-filesystem generated and maintained by the Linux kernel. Its contents represent live system and process information rather than ordinary persistent files stored on disk.

5. Why could /proc be useful during security investigation?

It can reveal running processes, process IDs, command-line information, open file descriptors, network-related information, memory and system state. Investigators can use it to identify unexpected or suspicious processes and their relationships to system resources.

Screenshot 14 – CPU and memory information

Run: cat /proc/cpuinfo; cat /proc/meminfo

```
(kali@kali) - [~/cyber_lab/reports]
$ cat /proc/cpuinfo
processor       : 0
vendor_id     : AuthenticAMD
cpu family    : 25
model         : 68
model name    : AMD Ryzen 7 170 with Radeon Graphics
stepping      : 1
microcode     : 0xffffffff
cpu MHz       : 3193.916
cache size    : 512 KB
physical id   : 0
siblings      : 2
core id       : 0
cpu cores     : 2
apicid        : 0
initial apicid : 0
fpu           : yes
fpu_exception : yes
cpuid level   : 13
wp            : yes
```

```

(kali@kali)-[~/cyber_lab/reports]
$ cat /proc/meminfo
MemTotal:      1978704 kB
MemFree:       117104 kB
MemAvailable:  800312 kB
Buffers:       40156 kB
Cached:        758620 kB
SwapCached:    23252 kB
Active:        757236 kB
Inactive:      714716 kB
Active(anon):  385064 kB
Inactive(anon): 323900 kB
Active(file):  372172 kB
Inactive(file): 390816 kB
Unevictable:   112 kB
Mlocked:       112 kB
SwapTotal:     976556 kB
SwapFree:      834368 kB
Zswap:         0 kB
Zswapped:      0 kB
Dirty:         0 kB
Writeback:     0 kB
AnonPages:     666740 kB
Mapped:        205016 kB
Shmem:         36028 kB

```

Screenshot 15 – Current shell PID and /proc/<PID>

Run: `echo $$; ls -ld /proc/$$`

```

(kali@kali)-[~/cyber_lab/reports]
$ echo $$
448937

(kali@kali)-[~/cyber_lab/reports]
$ ls -ld /proc/$$
dr-xr-xr-x 9 kali kali 0 Sep  8 13:39 /proc/448937

```

Exercise 15 – Explore /dev

Examine /dev/null and /dev/zero.

```
ls -l /dev/null
ls -l /dev/zero
file /dev/null
file /dev/zero
```

1. What does /dev represent?

/dev is a virtual filesystem containing device nodes and special interfaces used to interact with hardware devices and kernel-provided pseudo-devices.

2. Are the entries ordinary user files?

No. Many entries are device nodes or other special filesystem objects rather than ordinary regular files.

3. Determine the file type of /dev/null.

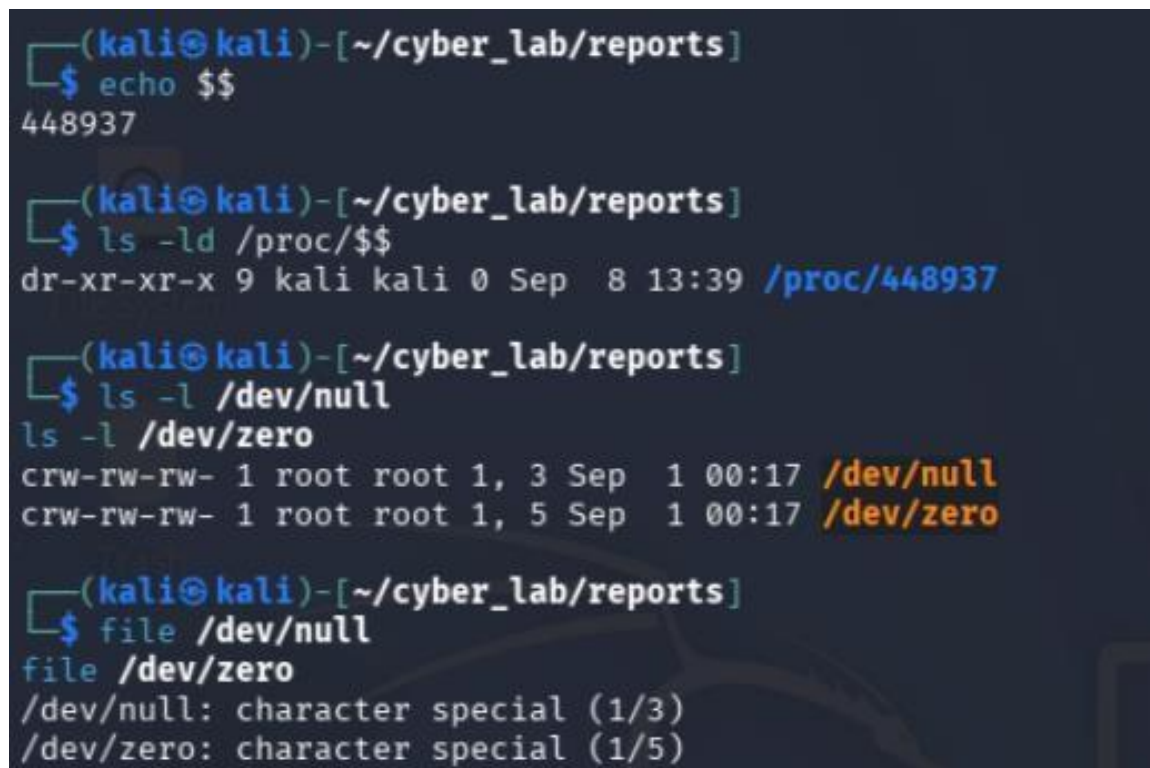
It is a character special device.

4. Why does Linux expose devices through the filesystem?

The filesystem interface provides a consistent way for programs to access devices and kernel interfaces using familiar file operations such as open, read and write.

Screenshot 16 – /dev/null and /dev/zero

Run: ls -l /dev/null /dev/zero; file /dev/null /dev/zero



```
(kali@kali)-[~/cyber_lab/reports]
$ echo $$
448937

(kali@kali)-[~/cyber_lab/reports]
$ ls -ld /proc/$$
dr-xr-xr-x 9 kali kali 0 Sep  8 13:39 /proc/448937

(kali@kali)-[~/cyber_lab/reports]
$ ls -l /dev/null
ls -l /dev/zero
crw-rw-rw- 1 root root 1, 3 Sep  1 00:17 /dev/null
crw-rw-rw- 1 root root 1, 5 Sep  1 00:17 /dev/zero

(kali@kali)-[~/cyber_lab/reports]
$ file /dev/null
file /dev/zero
/dev/null: character special (1/3)
/dev/zero: character special (1/5)
```

Exercise 16 – Filesystem Space Investigation

Determine total filesystem capacity, used space, available space, percentage utilization, home directory usage, and cyber_lab usage.

```
df -h
```

```
du -sh "$HOME"
```

```
du -sh "$HOME/cyber_lab"
```

```
# More detailed filesystem type information
```

```
df -T
```

For the filesystem containing your home directory, use `df -h "$HOME"`. The `Use%` column gives percentage utilization. `du -sh` reports the amount of disk space consumed by the specified directory tree.

Measurement	Command / Where to read
Total filesystem capacity	<code>df -h</code> — Size column
Used space	<code>df -h</code> — Used column
Available space	<code>df -h</code> — Avail column
Filesystem utilization	<code>df -h</code> — Use% column
Home directory space	<code>du -sh "\$HOME"</code>
cyber_lab space	<code>du -sh "\$HOME/cyber_lab"</code>

Explain the difference between `df` and `du`.

`df` reports filesystem-level space usage: how much capacity is allocated, used and available on a mounted filesystem. `du` estimates how much space is consumed by files and directories in a particular directory tree. Therefore, `df` is useful for overall filesystem capacity while `du` is useful for locating which directories consume space.

Screenshot 17 – `df` and `du` storage investigation

Run: `df -h`; `du -sh "$HOME"`; `du -sh "$HOME/cyber_lab"`

```

(kali@kali)~[~/cyber_lab/reports]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            869M   0 869M   0% /dev
tmpfs           194M  1.3M 193M   1% /run
/dev/sda1       79G   17G  58G  23% /
tmpfs           967M  4.0K 967M   1% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M  272K 966M   1% /tmp
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  116K 194M   1% /run/user/1000
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service

(kali@kali)~[~/cyber_lab/reports]
$ df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
udev            devtmpfs    889660         0    889660   0% /dev
tmpfs           tmpfs      197872      1236    196636   1% /run
/dev/sda1       ext4      82083148 17401336 60466264 23% /
tmpfs           tmpfs      989352         4    989348   1% /dev/shm
none            tmpfs       1024         0     1024   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs      989352      272    989080   1% /tmp
none            tmpfs       1024         0     1024   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs      197868      116    197752   1% /run/user/1000
none            tmpfs       1024         0     1024   0% /run/credentials/systemd-networkd.service

(kali@kali)~[~/cyber_lab/reports]
$ du -sh "$HOME"
du: cannot read directory '/home/kali/Linux/sample_dr': Permission denied
120M    /home/kali

(kali@kali)~[~/cyber_lab/reports]
$ du -sh "$HOME/cyber_lab"
32K     /home/kali/cyber_lab

```

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported. You are initially asked to examine only the filesystem. Identify which locations you would investigate and explain why.

Location	Why investigate it?	Security-relevant information
/home	User activity and personal data are stored here.	User files, scripts, downloads, shell history and configuration artifacts.
/etc	Contains important system and service configuration.	Accounts, authentication configuration, host/network configuration, service settings and persistence-related configuration.
/var/log	Contains system and application event records.	Authentication events, service activity, errors, kernel/system events and investigation timelines.
/tmp	Temporary files can contain transient artifacts.	Temporary scripts, downloaded/staged files, unusual ownership, timestamps and suspicious executables.
/proc	Provides live kernel and process information.	Running processes, PIDs, command-line/process details, open resources and live system state.
/dev	Contains device and special interfaces.	Device nodes and special interfaces; unusual device-related artifacts may warrant investigation.
/root	Root user's home directory may contain	Root shell configuration, scripts, command history where enabled, downloads and other

	administrative artifacts.	administrative files.
--	---------------------------	-----------------------

Commands used for the investigation

```
ls -la /home
ls -la /root
ls -lah /etc
ls -lah /var/log
ls -lah /tmp
ls -la /proc | head -30
ls -la /dev | head -30
```

These commands should be used in a read-only manner for the assignment. Avoid modifying, deleting, executing or changing permissions on system artifacts.

Screenshot 18 – Supplementary investigation evidence (Exercise 17)

Run: `ls -la /home; ls -la /root; ls -lah /etc; ls -lah /var/log; ls -lah /tmp; ls -la /proc | head -30`

```
total 24K
drwxr-xr-x  6 root    root    4.0K Sep  1 00:23 .
drwxr-xr-x 19 root    root    4.0K Sep  3 00:13 ..
drwx----- 5 alice   alice   4.0K Sep  3 00:34 alice
drwx----- 5 bob     bob     4.0K Sep  1 00:23 bob
drwx----- 5 charlie charlie 4.0K Sep  1 00:23 charlie
drwx--x--x 26 kali    kali    4.0K Sep  8 13:50 kali
ls: cannot open directory '/root': Permission denied
total 1.7M
drwxr-xr-x 186 root    root    12K Sep  7 23:43 .
drwxr-xr-x  19 root    root    4.0K Sep  3 00:13 ..
-rw-r--r--   1 root    root    3.8K Jun  1 01:06 adduser.conf
drwxr-xr-x   2 root    root    20K Aug 11 10:08 alternatives
drwxr-xr-x   8 root    root    4.0K Jun 16 10:30 apache2
drwxr-xr-x   2 root    root    4.0K Aug 11 10:07 apparmor
drwxr-xr-x   9 root    root    12K Aug 11 10:10 apparmor.d
drwxr-xr-x   8 root    root    4.0K Jun 16 11:05 apt
drwxr-xr-x   2 root    root    4.0K Jun 16 10:28 arp-scan
drwxr-xr-x   3 root    root    4.0K Jun 16 10:25 avahi
-rw-r--r--   1 root    root    2.5K May  3 15:28 bash.bashrc
-rw-r--r--   1 root    root    45 Sep 28 2025 bash_completion
drwxr-xr-x   2 root    root    4.0K Jun 16 10:23 bash_completion.d
-rw-r--r--   1 root    root    367 May  6 18:25 bindresvport.blacklist
drwxr-xr-x   2 root    root    4.0K Mar 23 09:27 binfmt.d
drwxr-xr-x   2 root    root    4.0K Aug 11 10:10 bluetooth
drwxr-xr-x   3 root    root    4.0K Jun 16 10:23 ca-certificates
-rw-r--r--   1 root    root    5.2K Jun 16 10:23 ca-certificates.conf
drwxr-s---   2 root    dip     4.0K Jun 16 10:25 chatscripts
drwxr-xr-x   2 root    root    4.0K Aug 11 10:10 chromium
drwxr-xr-x   2 root    root    4.0K Aug 11 10:10 chromium.d
drwxr-xr-x   2 root    root    4.0K Jun 16 10:28 cifs-utils
drwxr-xr-x   3 root    root    4.0K Jun 16 10:23 cloud
drwxr-xr-x   2 colord  colord  4.0K Aug 11 09:37 colord
drwxr-xr-x   2 root    root    4.0K Aug 11 09:29 console-setup
drwxr-xr-x   2 root    root    4.0K Aug 11 10:08 cracklib
drwx----- 2 root    root    4.0K Mar 23 09:27 credstore
drwx----- 2 root    root    4.0K Mar 23 09:27 credstore.encrypted
drwxr-xr-x   2 root    root    4.0K Jun 16 10:29 cron.d
drwxr-xr-x   2 root    root    4.0K Jun 16 10:30 cron.daily
drwxr-xr-x   2 root    root    4.0K Jun 16 10:23 cron.hourly
drwxr-xr-x   2 root    root    4.0K Jun 16 10:23 cron.monthly
```

```

-rw-r--r-- 1 root root root 8.6K Sep 8 13:39 vmware-vmtoolsd-root.log
-rw-r--r-- 1 root root kali 6.7K Sep 1 00:17 vmware-vmtoolsd-kali.log
-rw-r--r-- 1 root root root 7.7K Sep 1 00:16 vmware-vmtoolsd-root.log
-rw-r--r-- 1 root kali 17K Sep 1 00:17 vmware-vmtoolsd-kali.log
-rw-r--r-- 1 root utmp 11K Sep 1 00:17 wtmp
-rw-r--r-- 1 root root 8.0K Sep 3 01:45 wtmp.db
-rw-r--r-- 1 root root 35K Sep 8 13:40 Xorg.0.log
-rw-r--r-- 1 root root 28K Sep 1 00:15 Xorg.0.log.old
total 12K
drwxrwxrwt 13 root root 320 Sep 8 14:09 .
drwxr-xr-x 19 root root 4.0K Sep 3 00:13 ..
-rw-r--r-- 1 kali kali 0 Sep 1 00:17 config-err-0TKYgZ
drwxrwxrwt 2 root root 40 Sep 1 00:16 .font-unix
drwxrwxrwt 2 root root 60 Sep 1 00:17 .ICE-unix
drwxr-xr-x 3 root root 60 Sep 1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-colord.service
-bMcdDD
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-ModemManager.s
ervice-bxfVGP
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-polkit.service
-UYJ3xL
drwxr-xr-x 3 root root 60 Sep 1 00:16 systemd-private-45a3d694897f445c82c735328e937a09-systemd-logind
.service-DYQGuj
drwxr-xr-x 3 root root 60 Sep 1 00:17 systemd-private-45a3d694897f445c82c735328e937a09-upower.service
-xLbchu
drwxrwxrwt 2 root root 40 Sep 1 00:16 VMwareDnD
drwxr-xr-x 2 root root 40 Sep 1 00:16 vmware-root_464-826976206
-r--r--r-- 1 root root 11 Sep 1 00:17 .X0-lock
drwxrwxrwt 2 root root 60 Sep 1 00:17 .X11-unix
-rw-r--r-- 1 kali kali 398 Sep 1 00:17 .xfsm-ICE-HVV0U3
drwxrwxrwt 2 root root 40 Sep 1 00:16 .XIM-unix
ls: cannot access '/proc/465481': No such file or directory
ls: cannot access '/proc/465488': No such file or directory
ls: cannot access '/proc/465489': No such file or directory
total 4.0K
dr-xr-xr-x 304 root root 0 Sep 1 00:16 .
drwxr-xr-x 19 root root 4.0K Sep 3 00:13 ..
dr-xr-xr-x 9 root root 0 Sep 1 00:16 1
dr-xr-xr-x 9 root root 0 Sep 1 00:16 10
dr-xr-xr-x 9 root root 0 Sep 1 00:16 100
dr-xr-xr-x 9 root root 0 Sep 1 00:16 101
dr-xr-xr-x 9 root root 0 Sep 1 00:16 102
dr-xr-xr-x 9 root root 0 Sep 1 00:16 103

```